Gas Turbine Efficiency Laboratory

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1. Introduction

Gas turbines are the basis of aircraft propulsion. Their performance is described by thrust, thrust specific fuel consumption (TSFC), and the efficiencies of each component. The Brayton cycle is an effective thermodynamic model for these engines. The rig used in this laboratory is a small gas generator. It has a compressor and turbine on a common shaft, with a single gas fuelled combustor, producing thrust. All the instruments are calibrated at standard conditions so the variations to speeds, mass flows, and powers are corrected to standard conditions (288 K, 101.325 kPa) to effectively compare the tests.

1.1. Aim

The aim of the laboratory is to evaluate the performance of a thrust-producing gas turbine over generator speeds of 1000–1500 RPS. From this, we will determine the fuel-air ratio, compressor pressure ratio, net thrust, TSFC, and compressor-to-turbine power ratio against the corrected gas generator speed. We will then assess the burner efficiency at each speed using a first law energy balance across the combustor. This is done to build an understanding of the operating conditions of these types of engines.

1.2. Experiment description

The rig is a Cussons two-shaft unit, but the free power turbine is bypassed so that the exhaust from the gas generator turbine flows directly to a convergent thrust nozzle with exit area $A_9 = 0.00173$ m^2 (47 mm diameter). During the thermodynamic analysis a set of station numbers was adhered to. 1 is the compressor inlet, 2 for the compressor exit, 3 for the combustor exit / turbine inlet, 4 for the turbine exit, and 9 for the nozzle exit. We also applied the following assumptions:

- Measured values are total (stagnation) conditions due to low flow speeds.
- Zero inlet velocity
- Subsonic incompressible nozzle flow
- Ideal gas
- Cold side $\gamma = 1.4$, hot side γ is variable and determined by the handout chart, which gives $\gamma/(\gamma - 1) = c_\mathrm{p}/R$ against temperature. Hot-side c_p is evaluated at T_4 for the turbine and at the mean temperature $(T_2 + T_3)/2$ for the combustor.
- 100% mechanical efficiency
- Exit density uses a 0.6 correction factor on P_9 due to onset compressible effects, as given in the lab notice.

Additionally, propane fuel was used with an LHV of 46 MJ/kg. The fuel flow was read directly from the control panel scale in g/s and scaled by the correction factor given in the handout chart (Equation 8, Equation 9). For safety, hearing protection was worn throughout the test, an emergency shut-off button was noted on the panel, the exhaust fan was run to ventilate the test cell, and care was taken around the hot exhaust (above 500 °C).

The operating procedure was undertaken by the demonstrator, setting the RPS from 1000 to 1500 in steps of 100 RPS, and all readings were taken by our group once the engine was stable at each speed. We also recorded the actual speed and used this instead of the nominal speeds.

To measure the operating conditions a series of gauges were observed. The inlet air mass flow was read by a manometer reading h (mmH_2O) using $\dot{m}_a = 11.22\sqrt{h}$ g/s (Equation 6). Every station had a thermocouple, which was used to observe temperatures. The pressures came from a mixture of gauge and absolute gauges, and a mercury manometer for combustor pressure drop. The nozzle dynamic pressure (mbar) was measured which determines the exit velocity. Lastly, ambient pressure was assumed as standard (101.325 kPa) for all tests, and T_1 was taken to be the room temperature.

2. Results

Table 1 lists all the measured readings against N_{GG} and Table 2 lists the corrected and calculated quantities. Throughout all tests, the room temperature increased from 30 to 45 °C as the engine heated the testing room. As a result, the corrected speeds are lower than the recorded speeds and the corrected mass flows are higher than the actual mass flows (Equation 5, Equation 7, Equation 10). Due to this heating, the tests are not all evenly spaced in corrected speed. As a result, all plots and comparisons are drawn with respect to this corrected speed, including the thrust and TSFC calculations as required by the brief, with TSFC using the corrected fuel flow $\dot{m}_{f,corr}$ (Equation 18).

Table 1: Measured data for each test. P_a was not recorded and is taken as standard; P_2 was not read directly and is calculated from P_3 and $P_2 - P_3$ (Equation 2, Equation 3, Equation 4).

Quantity	**Units**	**Test 1**	**Test 2**	**Test 3**	**Test 4**	**Test 5**	**Test 6**
$N_{GG,nom}$	RPS	1000	1100	1200	1300	1400	1500
N_{GG} (actual)	RPS	1020	1100	1200	1320	1410	1515
T_a	°C	30	37	40	44	45	44
P_a	Pa	101325	101325	101325	101325	101325	101325
P_1	Pa	101325	101325	101325	101325	101325	101325
T_1	°C	30	37	40	44	45	44
Air flow, h	mmH_2O	147	152	161	173	182	192
T_2	°C	62	71	80	89	98	108
Fuel flow (scale)	g/s	1.25	1.37	1.45	1.70	1.90	2.20
T_g	°C	33	36	38	39	40	40
$P_2 - P_3$	cmHg	2.0	2.2	2.4	2.6	2.8	2.9
P_3	bar g	0.30	0.35	0.41	0.49	0.55	0.60
T_3	°C	610	620	650	680	700	730
P_4	bar abs	1.05	1.06	1.06	1.07	1.08	1.09
T_4	°C	480	490	500	515	520	540
P_9	bar abs	1.04	1.02	1.02	1.02	1.01	1.01
T_9	°C	460	465	480	490	510	530
Nozzle flow, ΔP_9	mbar	37.5	44	55	66	77	91

Table 2: Calculated results for each test.

Quantity	Units	Test 1	Test 2	Test 3	Test 4	Test 5	Test 6
$N_{\mathrm{GG,nom}}$	RPS	1000	1100	1200	1300	1400	1500
$N_{\mathrm{GG,corr}}$	RPS	994.2	1060	1151	1258	1342	1444
$\dot{m}_{\mathrm{a}}$	kg/s	0.1360	0.1383	0.1424	0.1476	0.1514	0.1555
$\dot{m}_{\mathrm{a,corr}}$	kg/s	0.1396	0.1436	0.1485	0.1549	0.1591	0.1631
$\dot{m}_{\mathrm{f}}$	kg/s	0.001072	0.001179	0.001251	0.001476	0.001657	0.001929
$\dot{m}_{\mathrm{f,corr}}$	kg/s	0.001100	0.001224	0.001305	0.001549	0.001742	0.002025
f	–	0.007884	0.008526	0.008787	0.01000	0.01095	0.01241
P_2	kPa	134.0	139.3	145.5	153.8	160.1	165.2
P_2/P_1	–	1.322	1.374	1.436	1.518	1.580	1.630
ρ_9	kg/m^3	0.2966	0.2889	0.2831	0.2794	0.2696	0.2629
V_9	m/s	159.0	174.5	197.1	217.3	239.0	263.1
F_{net}	N	21.80	24.35	28.31	32.40	36.57	41.41
TSFC	kg/N · hr	0.1817	0.1810	0.1659	0.1721	0.1714	0.1760
$\dot{W}_{\mathrm{c}}$	kW	4.373	4.724	5.720	6.671	8.059	9.995
$\dot{W}_{\mathrm{t}}$	kW	19.44	19.82	23.60	27.02	30.30	33.04
$\dot{W}_{\mathrm{c,corr}}$	kW	4.262	4.553	5.486	6.357	7.667	9.524
$\dot{W}_{\mathrm{t,corr}}$	kW	18.95	19.10	22.63	25.75	28.83	31.49
$\dot{W}_{\mathrm{c}}/\dot{W}_{\mathrm{t}}$	–	0.2249	0.2383	0.2424	0.2468	0.2660	0.3025
η_{b}	–	1.611	1.497	1.514	1.387	1.296	1.188

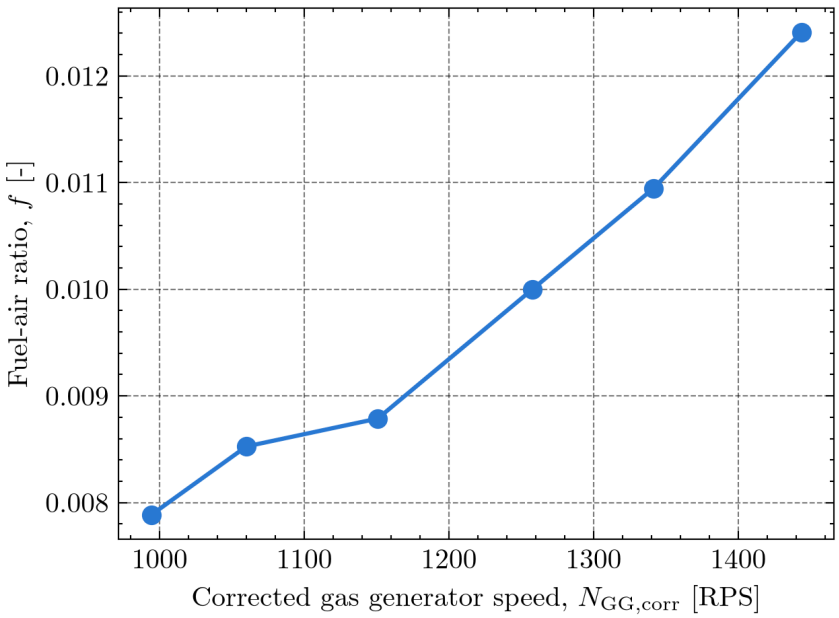


Figure 1: Fuel-air ratio against corrected gas generator speed.

The fuel-air ratio f rises steadily from 0.0079 to 0.0124 (+57%) with speed (Figure 1, Equation 11). This is an expected result since more fuel is needed to raise T_3 and the turbine work required to drive the compressor faster. The fuel flow rises 80% (1.07 to 1.93 g/s) while air flow rises only 14% (0.136 to 0.156 kg/s), so the trend in f arises almost entirely due to the fuel. The slope only steepens after Test 4, so these results tell us the demand is at least slightly faster than linear. Overall, the engine is very lean. Stoichiometric propane–air is $f \approx$

0.064, giving an equivalence ratio of only 0.12–0.19 (Equation 32, Equation 33). This is typical of gas turbines where most of the air is dilution air that keeps the turbine inlet temperature within material constraints.

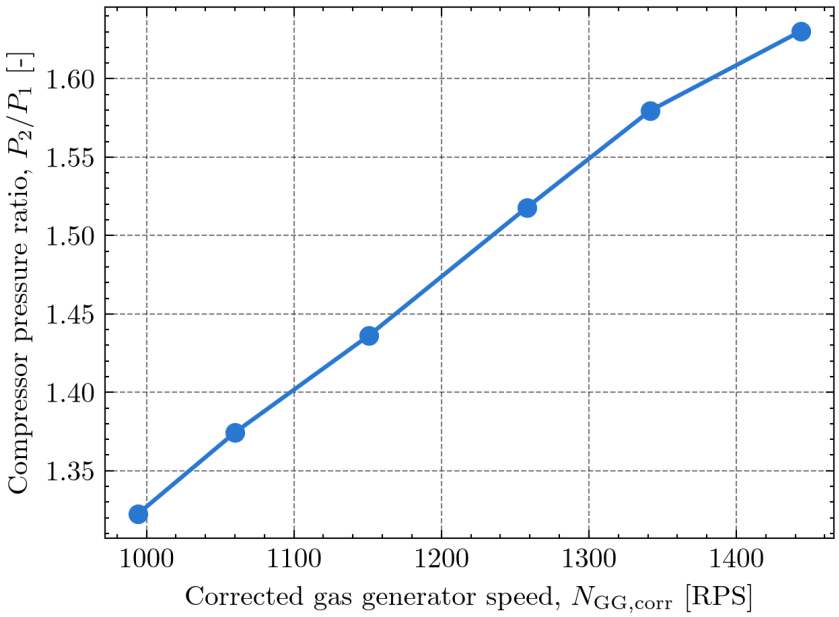


Figure 2: Compressor pressure ratio against corrected gas generator speed.

We see a similar trend here with the pressure ratio. Figure 2 shows the increase from 1.32 to 1.63 with speed (Equation 12). We know that the compressor pressure rise scales with blade speed squared, so for this narrow range of speeds the trend looks almost linear. The most significant deviation from this is the drop in increase rate between tests 5 and 6, which coincides with a drop in compressor isentropic efficiency (Equation 28). The efficiencies across the compressor are realistic, which suggests that the T_1, T_2, and P_2 readings are sound.

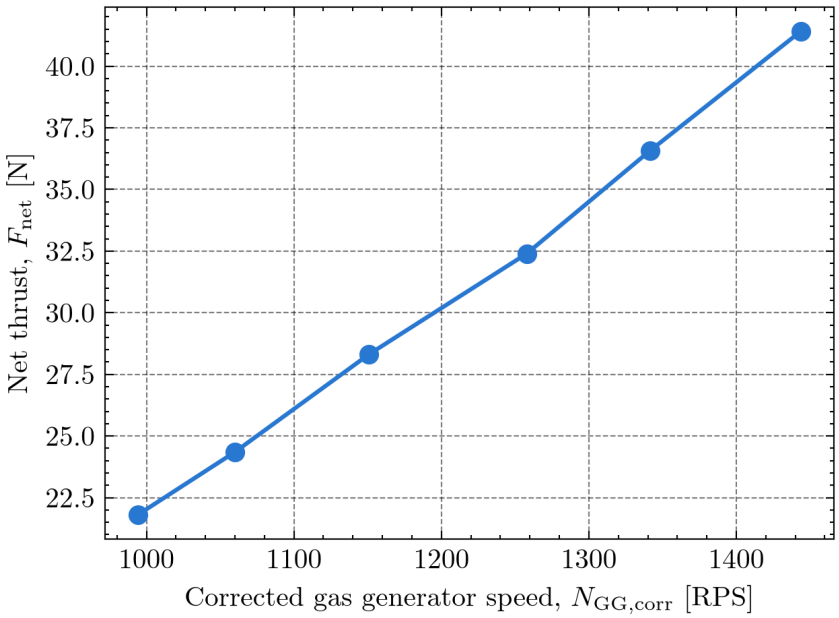


Figure 3: Net thrust against corrected gas generator speed.

We find that over the tests, the thrust nearly doubles from 21.8 to 41.4 N, a 90% increase (Equation 15, Equation 16). The increase mostly comes from the exit velocity (159 to 263 m/

s, +65%) rather than the mass flow (+15%), since $F = \dot{m}V_9$ and $V_9 \propto \sqrt{\Delta P_9}$. Since we have assumed zero inlet velocity, our thrust calculations do not include ram drag. Thus, the net thrust is the gross thrust. We also neglect pressure thrust since we have assumed that the convergent nozzle with subsonic exit flow must exit at ambient static pressure. We found that P_9 was within 0.7 kPa of ambient for tests 2–6, which is below the actual resolution of the pressure gauge (1 kPa). Test 1 is the only outlier.

The exit Mach number, however, is too high for the flow to be treated as fully incompressible. This is why a flat 0.6 correction factor was applied to the exit density (Equation 13), lowering ρ_9 and raising V_9 and F_{net} by about 29%. The velocity was determined from Bernoulli (Equation 14). If we instead used continuity, $V_9 = \dot{m}_9/\rho_9 A_9$ (Equation 34), the velocity would be 32–68% higher, with the largest gap at low speed. The lab notice states that the 0.6 factor should bring the two methods together, but for our data it widens the gap, since continuity scales with $1/\rho_9$ and Bernoulli with $1/\sqrt{\rho_9}$. Without the factor, continuity is only 2–30% above Bernoulli. The lab notice accepts either method, and Bernoulli was used as the additional notes suggest. It also relies only on the nozzle readings, whereas continuity also carries the error of the inlet air flow manometer.

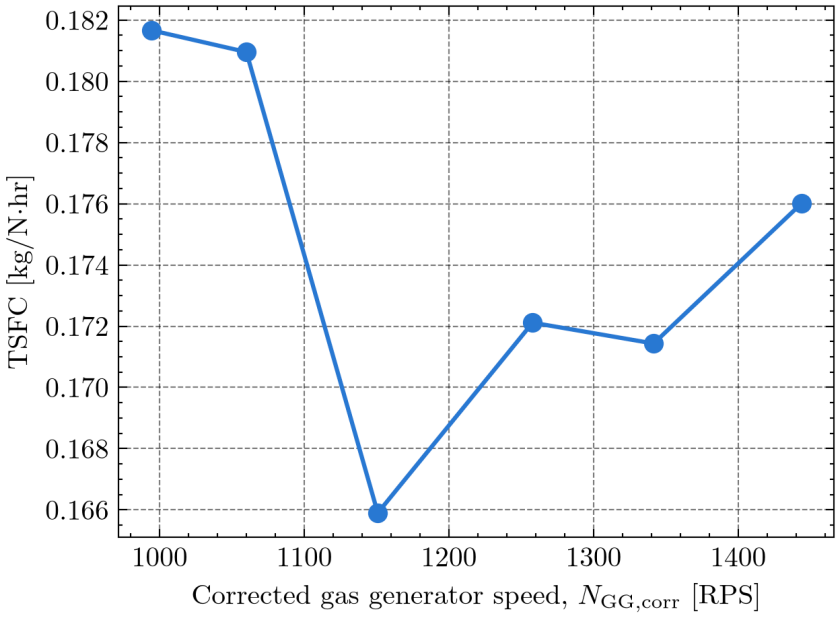


Figure 4: Thrust specific fuel consumption against corrected gas generator speed.

We found TSFC to be roughly flat at 0.17 – 0.18 kg/N·hr (Equation 18), with a minimum of 0.166 at 1151 RPS (Test 3). Fuel flow (+80%) and thrust (+90%) rise at similar rates so there is no strong trend. The observed scatter can be entirely attributed to the size expected from reading the fuel scale. Our observed TSFC is roughly double that of a typical turbojet, such as the GE J79 at 0.087 kg/N·hr at sea-level dry thrust[1], which would most likely be caused by our low pressure ratio and hence, low cycle efficiency.

[1]General Electric J79: 0.85 lb/(lbf·h) at military (dry) thrust. *General Electric J79*, Wikipedia, https://en.wikipedia.org/wiki/General_Electric_J79, accessed 25 October 2026.

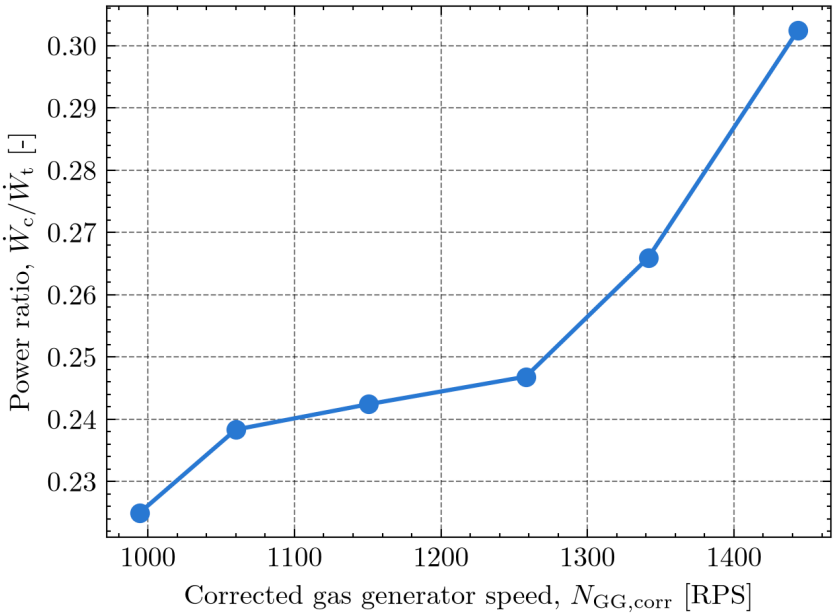


Figure 5: Ratio of compressor power to turbine power against corrected gas generator speed.

The compressor-to-turbine power ratio $\dot{W}_c/\dot{W}_t$ rises from 0.22 to 0.30 with speed (Figure 5; Equation 19, Equation 20, Equation 21, Equation 22, Equation 25). In a thrust turbine the gas generator turbine drives only the compressor through a shared shaft, so at a mechanical efficiency of 100% we would expect this to be about 1. As such, our calculated turbine power is 3–4.5 times too high. As mentioned previously, the compressor side looks credible. The compressor has a calculated isentropic efficiency of 0.74–0.89 (Equation 28), so the error is most likely on the turbine side. The measured $T_3 - T_4$ is 130–190 K, but only 29–57 K is needed to supply the compressor work (Equation 31). Additionally, the turbine isentropic efficiency implied by P_3/P_4 and T_3 would be 2.0–2.6 (Equation 30) which is impossible. So, either T_3 reads high (likely, explained more in further discussion), or there is a high heat loss in the uninsulated casing between stations 3 and 4. By the first law, $\dot{m}c_p(T_3 - T_4) = \dot{W}_t + \dot{Q}_{loss}$, so casing heat loss lowers T_4 below its adiabatic value and the apparent enthalpy drop overstates the shaft work delivered to the turbine blades. This inflates the apparent turbine power and also explains why the apparent isentropic efficiency $(T_3 - T_4)/(T_3 - T_{4s})$ exceeds 1. A high T_3 would also inflate burner efficiency.

We see the power ratio move towards 1 as speed increases, suggesting that the T_3 bias shrinks at higher flow (better mixing).

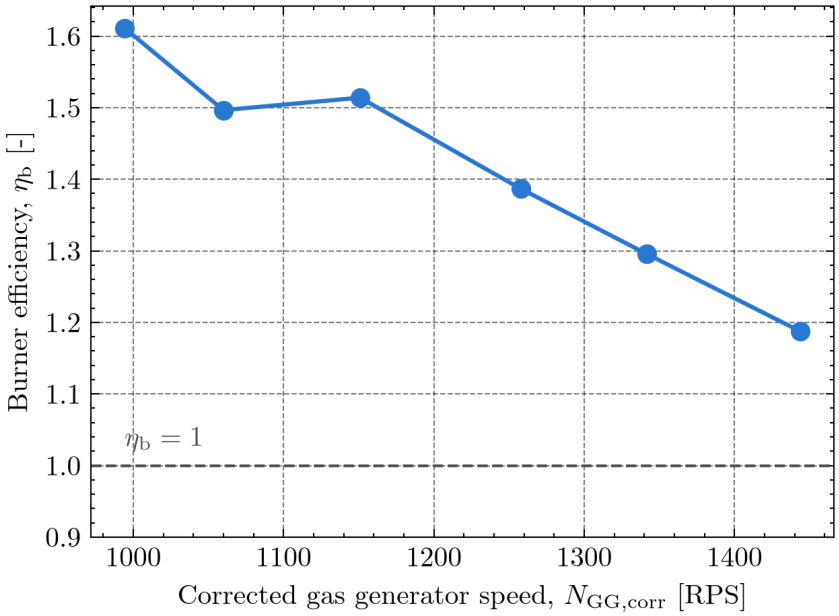


Figure 6: Burner efficiency against corrected gas generator speed.

Throughout the tests we see the burner efficiency fall from 1.61 to 1.19 with speed (Equation 26, Equation 27). A significant result of this analysis is that, in all tests, the calculated burner efficiency is consistently above 1. This violates the first law of thermodynamics. This means that the energy balance equation used to determine the efficiency is either overestimating the enthalpy rise across the combustor, or underestimating the fuel energy input. First, consider the possibility that the measured value for T_3 is overestimated. This is a highly likely cause since we know the engine uses a 'single gas fuelled tubular combustion chamber' (this is written on the physical control panel, see Figure 7 in the Appendix) and only has a single thermocouple probe at the combustor exit. Since the probe only measures the temperature at a single point rather than a mass-weighted average across the duct, the temperature reading will be heavily influenced by the unmixed hot core flow and radiation from the hot core.

However, T_3 is not the only possible cause. Even ignoring T_3 and looking at T_4 or T_9, where it is much more likely the gas mixture is fully mixed at these points, the thermal power carried by the exhaust is higher than the energy added by the fuel during combustion. Across the whole gas generator from station 1 to 9, the compressor work input and turbine work output cancel on the common shaft ($\dot{W}_c = \dot{W}_t$), and the measured stagnation temperature T_9 already includes the exhaust kinetic energy $V_9^2/2$, so the total enthalpy rise is $\dot{m}_9 c_p(T_{09} - T_{01}) = \dot{Q}_{fuel} - \dot{Q}_{loss}$. This already shows a violation of the first law. However, this only holds for the lower speed tests: the exhaust energy ratio (Equation 35) is 1.24–1.02 for Tests 1–4, but falls below 1 for Tests 5 and 6 (0.98 and 0.90). Hence, it is just as likely that the fuel-air ratio is underestimated, and that there was actually a greater fuel supply than calculated. This is a feasible cause since the control panel itself specifies to start with a supply pressure of 3 bar before reducing it to 1.5 bar before taking readings. If it was kept closer to 3 bar the whole run, the propane was denser than expected, so the true fuel mass flow is higher than what the scale on the control panel indicated. This is supported by the handout's fuel correction chart, which states that the rotameter scale reads g/s of propane at 15 °C and 1.5 kg/cm^2 g (about 1.5 bar g), with the correction curve given for 35 °C, whereas our fuel temperature T_g ranged from 33 to 40 °C. This could also explain how the exhaust energy exceeded the energy input from the fuel for tests 1–4. Heat is also lost to the test cell, which is why the inlet temperature must be monitored.

Hence, the true exhaust energy should be well below the fuel energy, meaning a ratio above 1 is very telling.

The trend towards 1 at higher speed matches the power ratio trend, which could indicate a common cause. Both the T_3 bias and the relative fuel scale reading error are the largest at low flows.

There are also some other smaller sources of error to consider, such as reading c_p from a chart, and assuming an LHV for pure propane. Real combustors have $\eta_b \approx 0.98$–0.99, so a reasonable result would sit just below 1.

3. Conclusion

Over the corrected speeds of 994–1444 RPS the thrust increased from 22 to 41 N, the pressure ratio from 1.32 to 1.63, f from 0.0079 to 0.0124, and TSFC roughly constant at approximately 0.17 kg/N·hr. The trends in f, pressure ratio, thrust, and TSFC make physical sense. Thrust is driven mainly by exit velocity, and our TSFC is high since our pressure ratio is low. We found, however, that both the power ratio (0.22–0.30) and burner efficiency (1.19–1.61) are not physically possible. Both issues point to an overestimated T_3 which is supported by the impossible turbine efficiency, and possibly an under-read fuel flow. Some recommendations for future tests would be to include a rake of thermocouples at the combustor exit to take a mass-averaged T_3 to mitigate the effects of radiation and unmixed gas (even better, radiation-shielded probes), insulate the turbine and ducting, log the fuel supply pressure, measure ambient pressure, and allow a longer stabilisation period as the engine apparatus heats up.

Appendix

A. Sample Calculations at Test 3

Sample calculations are shown for Test 3 (nominal $N_{\mathrm{GG}} = 1200$ RPS).

A.1. Unit conversions

Room temperature:

$$\begin{aligned} T_{\mathrm{a}} &= T_{\mathrm{a},^\circ\mathrm{C}} + 273.15 \\ &= 40 + 273.15 = 313.15\,\mathrm{K} \end{aligned} \tag{1}$$

Turbine inlet pressure (gauge to absolute):

$$\begin{aligned} P_3 &= P_{3,\text{bar g}} \times 10^5 + P_{\mathrm{a}} \\ &= 0.41 \times 10^5 + 101325 = 142325\,\mathrm{Pa} \end{aligned} \tag{2}$$

Combustor pressure drop:

$$\begin{aligned} \Delta P_{23} &= \Delta P_{23,\mathrm{cmHg}} \times 1333.22 \\ &= 2.4 \times 1333.22 = 3199.73\,\mathrm{Pa} \end{aligned} \tag{3}$$

Compressor exit pressure:

$$\begin{aligned} P_2 &= P_3 + \Delta P_{23} \\ &= 142325 + 3199.73 = 145524.73\,\mathrm{Pa} \end{aligned} \tag{4}$$

A.2. Corrections and standardisation

Corrected gas generator speed:

$$\begin{aligned} N_{\mathrm{GG,corr}} &= N_{\mathrm{GG}}\sqrt{\frac{288}{T_{\mathrm{a}}}} \\ &= 1200\sqrt{\frac{288}{313.15}} = 1150.8\,\mathrm{RPS} \end{aligned} \tag{5}$$

Actual air mass flow:

$$\begin{aligned} \dot{m}_{\mathrm{a}} &= \frac{11.22\sqrt{h}}{1000} \\ &= \frac{11.22\sqrt{161}}{1000} = 0.1424\,\mathrm{kg/s} \end{aligned} \tag{6}$$

Corrected air mass flow:

$$\begin{aligned} \dot{m}_{\mathrm{a,corr}} &= \dot{m}_{\mathrm{a}} \times \frac{1.01325}{P_{\mathrm{a,bar}}} \times \sqrt{\frac{T_{\mathrm{a}}}{288}} \\ &= 0.1424 \times \frac{1.01325}{1.01325} \times \sqrt{\frac{313.15}{288}} = 0.1485\,\mathrm{kg/s} \end{aligned} \tag{7}$$

Fuel correction factor (from the handout graph):

$$\mathrm{CF} = \mathrm{CF}(1.45\,\mathrm{g/s}) = 0.8628 \tag{8}$$

Actual fuel mass flow:

$$\begin{aligned}\dot{m}_{\mathrm{f}} &= \frac{\dot{m}_{\mathrm{f,measured}} \times \mathrm{CF}}{1000} \\ &= \frac{1.45 \times 0.8628}{1000} = 0.001251\,\mathrm{kg/s}\end{aligned} \tag{9}$$

Corrected fuel mass flow:

$$\begin{aligned}\dot{m}_{\mathrm{f,corr}} &= \dot{m}_{\mathrm{f}} \times \frac{1.01325}{P_{\mathrm{a,bar}}} \times \sqrt{\frac{T_{\mathrm{a}}}{288}} \\ &= 0.001251 \times \frac{1.01325}{1.01325} \times \sqrt{\frac{313.15}{288}} = 0.001305\,\mathrm{kg/s}\end{aligned} \tag{10}$$

A.3. Performance parameters

Fuel-air ratio:

$$\begin{aligned}f &= \frac{\dot{m}_{\mathrm{f,corr}}}{\dot{m}_{\mathrm{a,corr}}} = \frac{\dot{m}_{\mathrm{f}}}{\dot{m}_{\mathrm{a}}} \\ &= \frac{0.001305}{0.1485} = 0.008787\end{aligned} \tag{11}$$

Compressor pressure ratio:

$$\begin{aligned}r_{\mathrm{c}} &= \frac{P_2}{P_1} \\ &= \frac{145524.73}{101325} = 1.436\end{aligned} \tag{12}$$

Nozzle exit density (ideal gas, with the 0.6 correction factor on P_9 from the lab notice):

$$\begin{aligned}\rho_9 &= \frac{0.6 P_9}{R T_9} \\ &= \frac{0.6 \times 1.02 \times 10^5}{287 \times (480 + 273.15)} = 0.2831\,\mathrm{kg/m^3}\end{aligned} \tag{13}$$

Nozzle exit velocity (Bernoulli):

$$\begin{aligned}V_9 &= \sqrt{\frac{2\Delta P_9}{\rho_9}} \\ &= \sqrt{\frac{2 \times 5500}{0.2831}} = 197.11\,\mathrm{m/s}\end{aligned} \tag{14}$$

Nozzle mass flow:

$$\begin{aligned}\dot{m}_9 &= \dot{m}_{\mathrm{a}} + \dot{m}_{\mathrm{f}} \\ &= 0.1424 + 0.001251 = 0.1436\,\mathrm{kg/s}\end{aligned} \tag{15}$$

Net thrust (static, no pressure thrust):

$$\begin{aligned}F_{\mathrm{net}} &= \dot{m}_9 V_9 \\ &= 0.1436 \times 197.11 = 28.31\,\mathrm{N}\end{aligned} \tag{16}$$

Corrected net thrust:

$$\begin{aligned} F_{\text{net,corr}} &= F_{\text{net}} \times \frac{1.01325}{P_{\text{a,bar}}} \\ &= 28.31 \times \frac{1.01325}{1.01325} = 28.31\,\text{N} \end{aligned} \tag{17}$$

Here F_{net} is evaluated from the actual nozzle mass flow $\dot{m}_9$ (Equation 15) and corrected for ambient pressure only, $F_{\text{net,corr}} = F_{\text{net}} \times 1.01325/P_{\text{a,bar}}$, while TSFC uses the corrected fuel flow $\dot{m}_{\text{f,corr}}$ (Equation 10). If F_{net} were instead evaluated from the corrected mass flow $\dot{m}_{9,\text{corr}} = \dot{m}_{\text{a,corr}} + \dot{m}_{\text{f,corr}}$, it would be higher by $\sqrt{T_{\text{a}}/288}$ (about 3–5%, e.g. 29.53 N instead of 28.31 N for Test 3), and TSFC would be lower by the same factor.

Thrust specific fuel consumption:

$$\begin{aligned} \text{TSFC} &= \frac{\dot{m}_{\text{f,corr}}}{F_{\text{net,corr}}} \times 3600 \\ &= \frac{0.001305}{28.31} \times 3600 = 0.1659\,\text{kg/N·hr} \end{aligned} \tag{18}$$

A.4. Compressor and turbine power

Cold specific heat:

$$\begin{aligned} c_{\text{p,cold}} &= \frac{\gamma R}{\gamma - 1} \\ &= \frac{1.4 \times 287}{1.4 - 1} = 1004.5\,\text{J/kg·K} \end{aligned} \tag{19}$$

Hot specific heat (from the handout graph at $T_4 = 773.15$ K):

$$\begin{aligned} c_{\text{p,hot}} &= R\left[\frac{\gamma}{\gamma - 1}\right]_{T_4} \\ &= 287 \times 3.816 = 1095.31\,\text{J/kg·K} \end{aligned} \tag{20}$$

Compressor power:

$$\begin{aligned} \dot{W}_{\text{c}} &= \dot{m}_{\text{a}} c_{\text{p,cold}} (T_2 - T_1) \\ &= 0.1424 \times 1004.5 \times (353.15 - 313.15) = 5720.26\,\text{W} \end{aligned} \tag{21}$$

Turbine power:

$$\begin{aligned} \dot{W}_{\text{t}} &= (\dot{m}_{\text{a}} + \dot{m}_{\text{f}}) c_{\text{p,hot}} (T_3 - T_4) \\ &= 0.1436 \times 1095.31 \times (923.15 - 773.15) = 23595.65\,\text{W} \end{aligned} \tag{22}$$

Corrected compressor power:

$$\begin{aligned} \dot{W}_{\text{c,corr}} &= \dot{W}_{\text{c}} \times \frac{1.01325}{P_{\text{a,bar}}} \times \sqrt{\frac{288}{T_{\text{a}}}} \\ &= 5720.26 \times \frac{1.01325}{1.01325} \times \sqrt{\frac{288}{313.15}} = 5485.75\,\text{W} \end{aligned} \tag{23}$$

Corrected turbine power:

$$\begin{aligned}\dot{W}_{\mathrm{t,corr}} &= \dot{W}_{\mathrm{t}} \times \frac{1.01325}{P_{\mathrm{a,bar}}} \times \sqrt{\frac{288}{T_{\mathrm{a}}}} \\ &= 23595.65 \times \frac{1.01325}{1.01325} \times \sqrt{\frac{288}{313.15}} = 22628.3\,\mathrm{W}\end{aligned} \tag{24}$$

Compressor to turbine power ratio:

$$\frac{\dot{W}_{\mathrm{c,corr}}}{\dot{W}_{\mathrm{t,corr}}} = \frac{\dot{W}_{\mathrm{c}}}{\dot{W}_{\mathrm{t}}} = \frac{5485.75}{22628.3} = 0.2424 \tag{25}$$

A.5. Burner efficiency

Hot specific heat (from the handout graph at $(T_2 + T_3)/2 = 638.15$ K):

$$\begin{aligned}c_{\mathrm{p,b}} &= R\left[\frac{\gamma}{\gamma - 1}\right]_{(T_2+T_3)/2} \\ &= 287 \times 3.708 = 1064.3\,\mathrm{J/kg{\cdot}K}\end{aligned} \tag{26}$$

Burner efficiency (first law energy balance, station 2 to 3; the actual f is used, which equals the corrected f from Equation 11):

$$\begin{aligned}\eta_{\mathrm{b}} &= \frac{(1+f)c_{\mathrm{p,b}}(T_3 - T_2)}{f \cdot \mathrm{LHV}} \\ &= \frac{(1 + 0.008787) \times 1064.3 \times (923.15 - 353.15)}{0.008787 \times 46 \times 10^6} = 1.514\end{aligned} \tag{27}$$

A.6. Discussion quantities

Compressor isentropic efficiency:

$$\begin{aligned}\eta_{\mathrm{c}} &= \frac{T_1\left[(P_2/P_1)^{(\gamma-1)/\gamma} - 1\right]}{T_2 - T_1} \\ &= \frac{313.15[1.436^{0.2857} - 1]}{353.15 - 313.15} = 0.8531\end{aligned} \tag{28}$$

Hot ratio of specific heats:

$$\begin{aligned}\gamma_{\mathrm{hot}} &= \frac{c_{\mathrm{p,hot}}}{c_{\mathrm{p,hot}} - R} \\ &= \frac{1095.31}{1095.31 - 287} = 1.355\end{aligned} \tag{29}$$

Turbine isentropic efficiency:

$$\begin{aligned}\eta_{\mathrm{t}} &= \frac{T_3 - T_4}{T_3\left[1 - (P_4/P_3)^{(\gamma_{\mathrm{hot}}-1)/\gamma_{\mathrm{hot}}}\right]} \\ &= \frac{923.15 - 773.15}{923.15[1 - (106000/142325)^{0.2620}]} = 2.187\end{aligned} \tag{30}$$

Turbine temperature drop needed to drive the compressor:

$$\begin{aligned}\Delta T_{\mathrm{t,req}} &= \frac{\dot{W}_{\mathrm{c}}}{(\dot{m}_{\mathrm{a}}+\dot{m}_{\mathrm{f}})c_{\mathrm{p,hot}}} \\ &= \frac{5720.26}{0.1436\times 1095.31} = 36.36\,\mathrm{K}\end{aligned} \tag{31}$$

Stoichiometric fuel-air ratio, from $C_3H_8 + 5(O_2 + 3.76\ N_2)$:

$$f_{\mathrm{st}} = \frac{M_{\mathrm{C_3H_8}}}{5\times 4.76\times M_{\mathrm{air}}} = \frac{44.097}{5\times 4.76\times 28.97} = 0.06396 \tag{32}$$

Equivalence ratio:

$$\varphi = \frac{f}{f_{\mathrm{st}}} = \frac{0.008787}{0.06396} = 0.1374 \tag{33}$$

Nozzle exit velocity (continuity):

$$\begin{aligned}V_{9,\mathrm{cont}} &= \frac{\dot{m}_{\mathrm{a}}+\dot{m}_{\mathrm{f}}}{\rho_9 A_9} \\ &= \frac{0.1436}{0.2831\times 0.00173} = 293.2\,\mathrm{m/s}\end{aligned} \tag{34}$$

Exhaust energy ratio (first law from inlet to nozzle exit, $c_{\mathrm{p,ex}}$ from the handout graph at $(T_1 + T_9)/2 = 533.15$ K):

$$\begin{aligned}\eta_{\mathrm{ex}} &= \frac{(1+f)c_{\mathrm{p,ex}}(T_9 - T_1)}{f\cdot \mathrm{LHV}} \\ &= \frac{(1+0.008787)\times 1041.86\times(753.15-313.15)}{0.008787\times 46\times 10^6} = 1.144\end{aligned} \tag{35}$$

B. Gas Turbine Control Panel

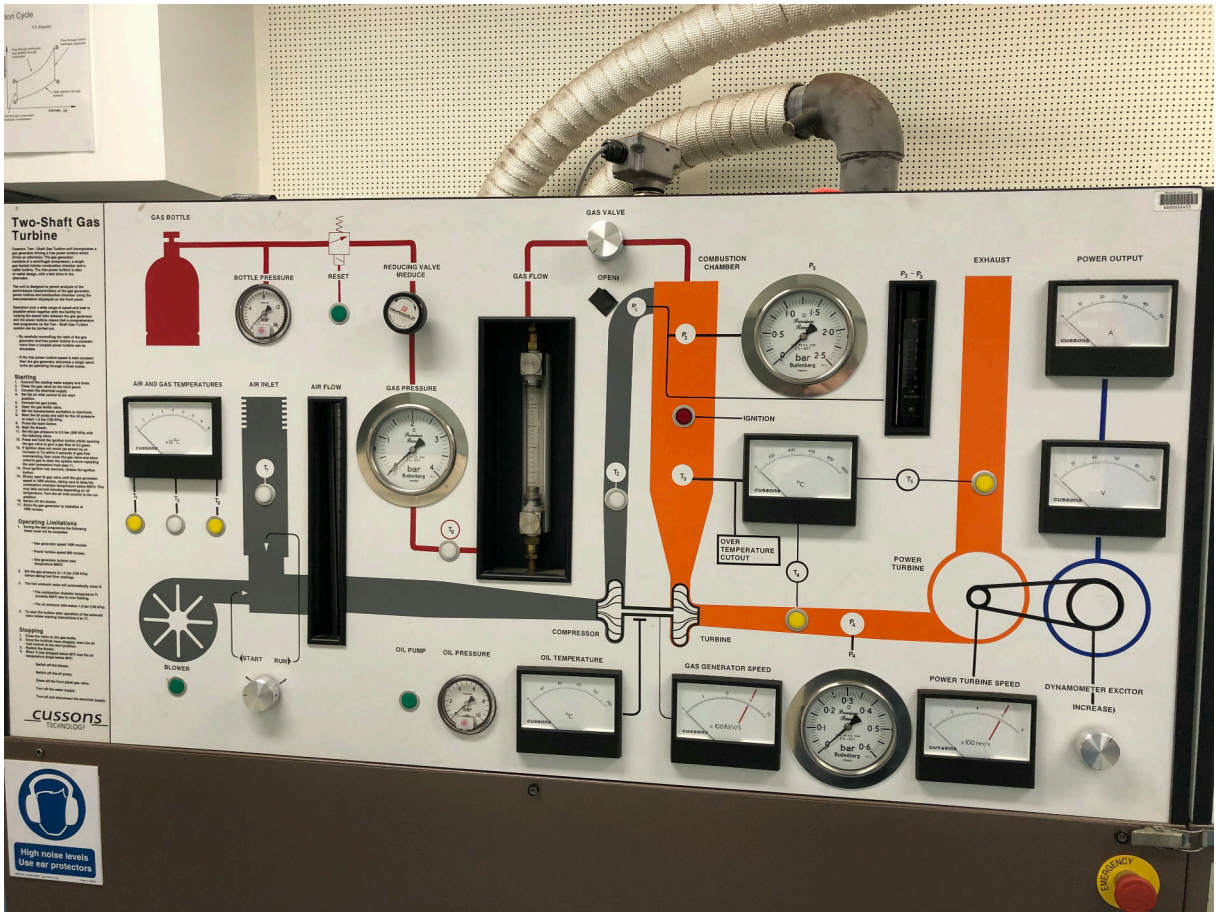


Figure 7: Control panel of the Cussons two-shaft gas turbine rig used in this laboratory. The panel description states that the gas generator uses a single gas fuelled tubular combustion chamber, and only one thermocouple (T_3) is fitted at the combustor exit.